

Sarthak Roy

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SUMMARY

AI/ML engineer specializing in production computer vision and document intelligence - real-time detection and tracking, feature matching, sensor fusion, and geometry-driven OCR reconstruction - building end-to-end Python systems from model inference through FastAPI deployment, with applied RAG and LLM experience.

EXPERIENCE

AI Engineer Trainee - Tardid Technologies Pvt. Ltd.

Bangalore | Feb 2026 - Aug 2026

Brainbox AI product suite - ML/CV modules across autonomous surveillance vessel, AI perimeter security, and networked warfare management platforms.

- **Document intelligence (OCR):** Engineered a geometry-first OCR reconstruction pipeline (PaddleOCR/PP-OCR) that converts fragmented text detections into structured multi-column ledger data using bounding-box geometry, spatial row clustering, column-interval assignment, and multi-line header detection.
- **OCR data quality:** Implemented character-confusion handling, confidence-aware filtering, and cell-level numeric/text normalization for noisy scans; extended the pipeline to multi-page documents with structured Excel output, validated by synthetic multi-page test suites.
- **Real-time detection & tracking:** Built vessel detection and tracking (YOLO + OC-SORT) with motion-gating and per-track centroid history to suppress false positives on continuously running RTSP video pipelines.
- **Feature matching & deployment:** Architected a vessel identification pipeline using LoFTR correspondences and DINOv2 descriptors with quality-adaptive thresholds and RANSAC geometric filtering, shipped as a FastAPI service with an operator review frontend.
- **Classical CV & sensor fusion:** Designed a dual-EMA background-subtraction intrusion/unattended-object detector with global lighting-change suppression to cut false alerts on live CCTV; fused PTZ camera and LiDAR data in ROS2, resolving range/bearing accuracy issues in multi-domain tracking.

Computer Vision Intern - Tardid Technologies Pvt. Ltd.

Bangalore | Aug 2023 - Oct 2023

- **Maritime detection under degraded conditions:** Engineered CNN-based ship detection and classification with image/video enhancement pipelines to sustain accuracy under fog and occlusion; investigated SPAD and acousto-optic deflection for precision imaging, and integrated perception models with naval hardware and data pipelines.
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PERSONAL PROJECTS

ParcelPilot - AI Support Operations Platform (REST APIs, OpenAPI, Automated Testing)

- Engineered a document-grounded AI support platform end to end: indexed operational documentation for retrieval and implemented a deterministic provider mode so AI responses stay reproducible and testable rather than free-running LLM output.
- Architected the backend around a validated OpenAPI contract with a full automated test suite (564 backend, 81 frontend tests passing), and deployed the integrated frontend and API as a working system.

MedBot - Medical Knowledge Assistant (RAG) (Python, LangChain, Pinecone, FastAPI, OpenRouter)

- Built an end-to-end retrieval-augmented generation pipeline: chunked and embedded medical PDFs into a Pinecone vector index, orchestrated context retrieval with LangChain, and served grounded LLM responses through Flask/FastAPI endpoints.

SubSpace - AI Chat & Authentication Platform (React.js, Supabase/Nhost, n8n, OpenRouter)

- Developed a full-stack chat platform with session-validated authentication, an OpenRouter-backed AI chat service, and n8n-driven real-time backend workflows; reduced onboarding time by 60%.
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TECHNICAL SKILLS

Languages: Python, JavaScript, SQL

Computer Vision: OpenCV, YOLO, OC-SORT, DINOv2, LoFTR, PaddleOCR/PP-OCR, Object Detection & Multi-Object Tracking, Feature Matching, RANSAC, Image & Document Processing, Table/Text Reconstruction, RTSP Pipelines, LiDAR-Camera Fusion, ROS2

AI / ML: Deep Learning, Convolutional Neural Networks (CNNs), Model Inference Pipelines, Retrieval-Augmented Generation (RAG), LangChain, Vector Embeddings, Vector Search, Hugging Face, OpenRouter

Backend & APIs: FastAPI, Flask, REST API Design, OpenAPI, WebSockets, Node.js, Express.js, n8n

Data & Platforms: PostgreSQL, MongoDB, Pinecone, Supabase, AWS, Git, React.js, Next.js

EDUCATION

B.Tech, Computer Science Engineering

Vellore Institute of Technology, Vellore | 2021 - 2025